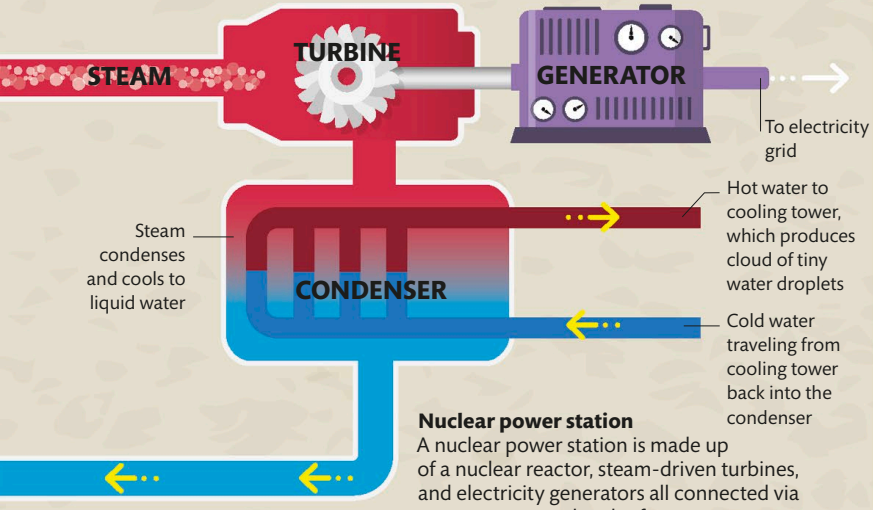
5 Turning the turbine
Housed in a turbine hall, the turbine's fan blades are turned by the high-pressure steam from the heat exchanger. The turbines typically spin at speeds of 1,800–3,600 rpm.

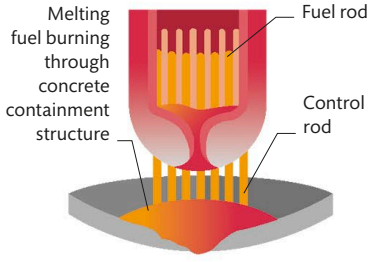
6 Power supply
The electricity generator is driven by the turbine's driveshaft. A transformer increases the electricity's voltage so that it can be carried by a local or regional power grid efficiently.



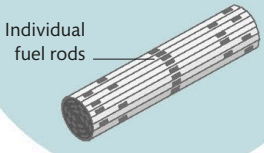
Nuclear power station
A nuclear power station is made up of a nuclear reactor, steam-driven turbines, and electricity generators all connected via numerous control and safety systems.

NUCLEAR MELTDOWN

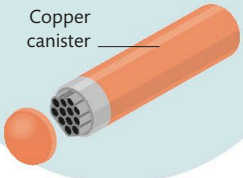
The failure of a nuclear reactor's coolant system can lead to excessive amounts of heat building up in the fuel rods. In extreme cases, the rods can melt and burn through their containment structure. This can release huge amounts of radioactivity that may contaminate the environment. In 2011, following an earthquake and tsunami, three of the reactors in Japan's Fukushima Daiichi plant suffered partial meltdowns.



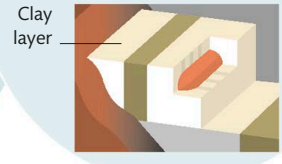
1 Fuel bundle
Spent fuel rods that emit high levels of heat and radioactivity are left to cool for several years.



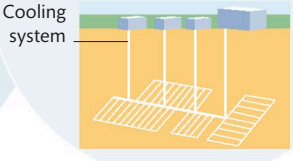
2 Disposed canister
Radioactive waste is vitrified by mixing it with inert molten glass. The mixture solidifies within canisters or capsules.



3 Sealed with clay
The capsules are buried surrounded by a thick layer of impermeable clay acting as an additional barrier.



4 Burial site
The containment burial site, 1,600–3,330 ft (500–1,000 m) below Earth's surface, is monitored and maintained.



Radioactive waste disposal

Used (or spent) fuel rods are removed from a reactor every 2–5 years but still generate heat for decades and emit harmful levels of radioactivity for even longer. Most are initially stored in deep, cold-water storage pools for a number of years before they are either reprocessed or placed in concrete-encased casks. Several countries have proposed plans to store waste deep underground, but no sites are yet operational.

Geological repository plans
One proposed solution for radioactive waste disposal involves the already established technique of vitrification followed by burial in temperature-regulated deep bore holes.

A 1,000 MW NUCLEAR PLANT PRODUCES ABOUT 30 TONS (27 TONNES) OF SPENT NUCLEAR FUEL A YEAR